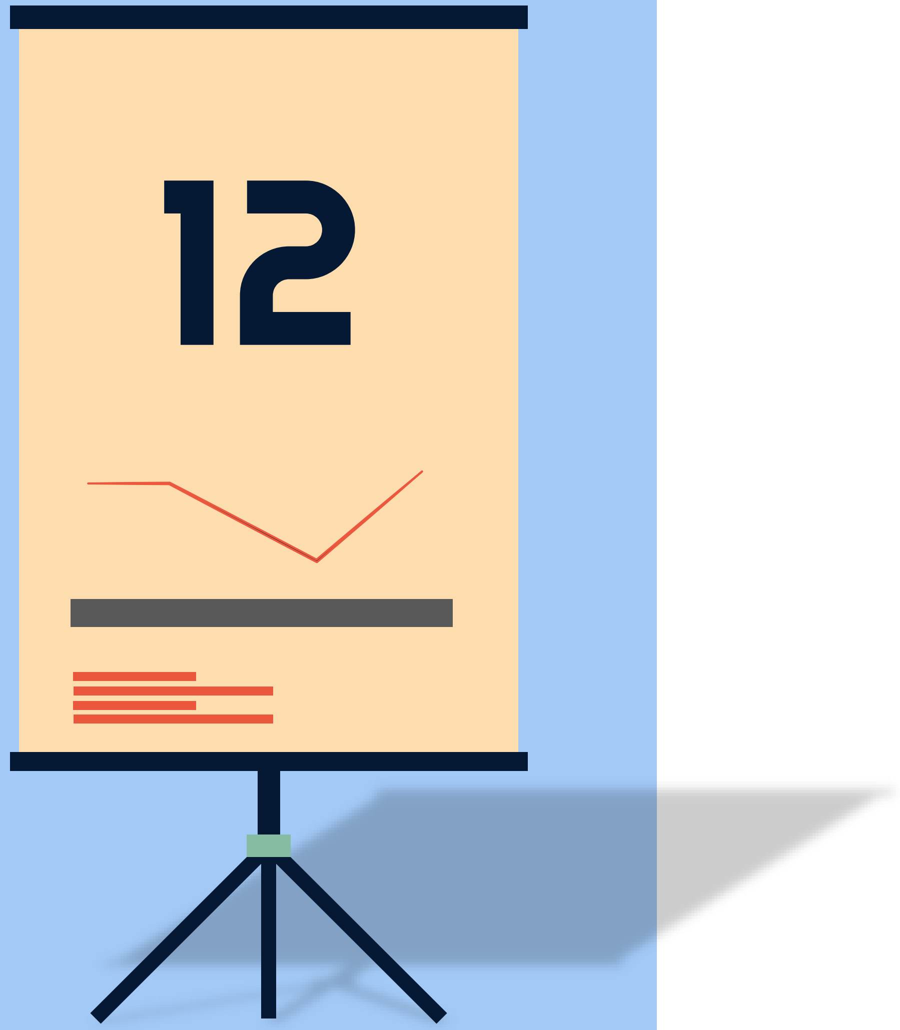
# Program Studi Statistika

Fakultas Matematika dan Ilmu Pengetahuan Alam

Universitas Negeri Makassar

# Ekonometrika

## 2025/2026




# Regresi Variabel Respons Kualitatif

# 1. Konsep Dasar

- Variabel penyusun model regresi dapat berupa variabel numerik dan juga variabel kategorik.

Numerik/
kategorik

$$Y = \beta X$$

- Pertemuan ini akan membahas variabel respons yang terdiri dari variabel kategori

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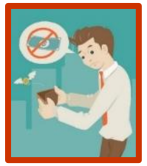
# 2 . Regresi Logistik

- Model regresi dengan variabel respons berupa variabel kategori disebut regresi logistik.
- Variabel respons bersifat biner/dikotomi, sehingga hanya ada 1 nilai dari dua kemungkinan.

$$Y = \begin{cases} 1 \\ 0 \end{cases}$$



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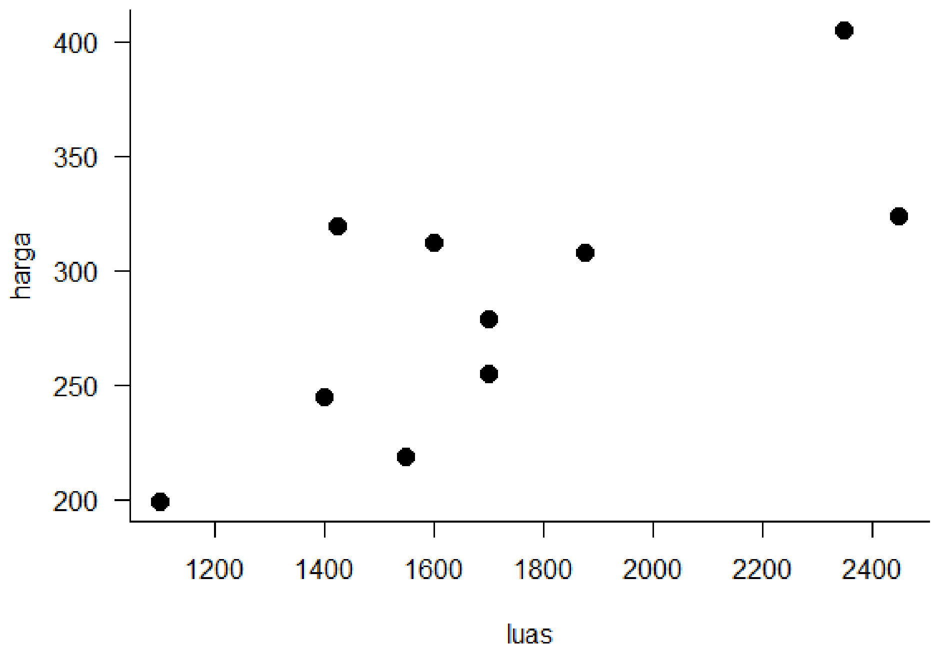
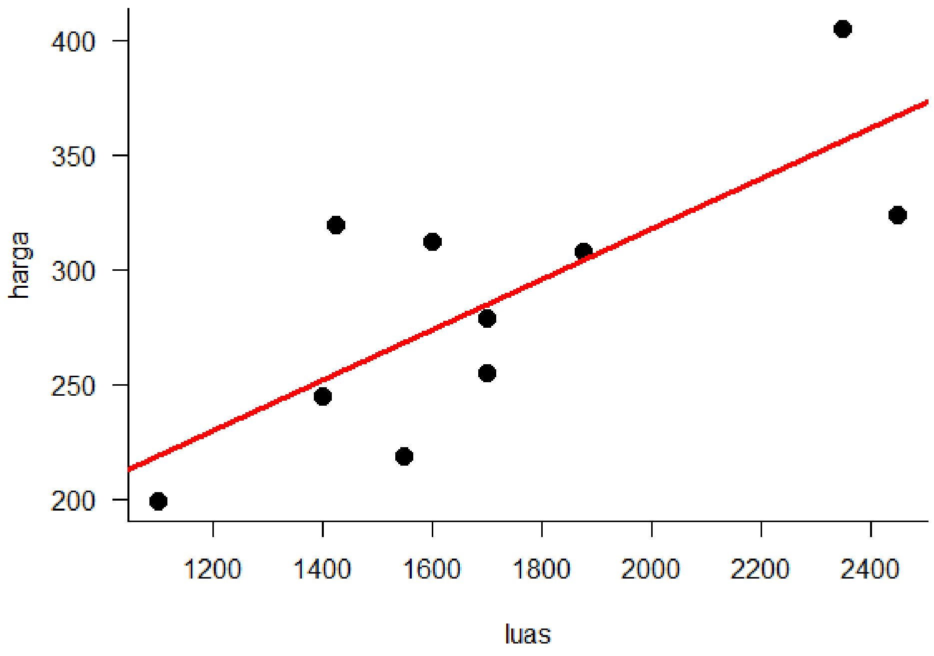
# Regresi Logistik: Permasalahan Umum

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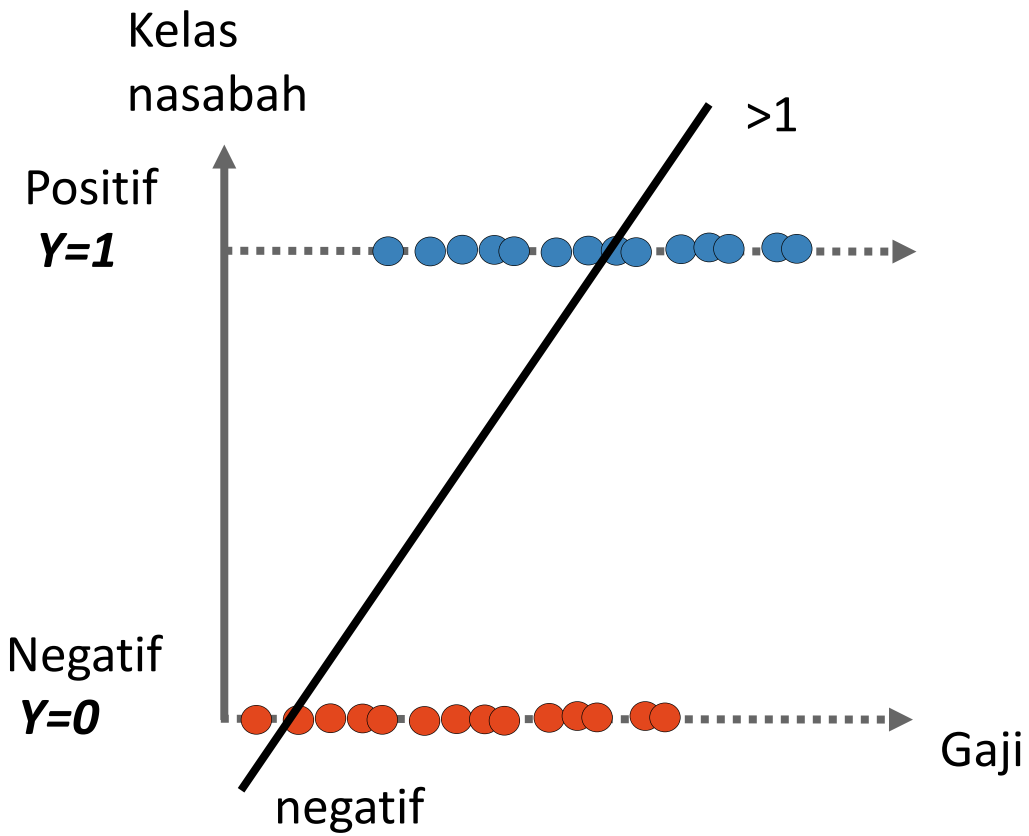
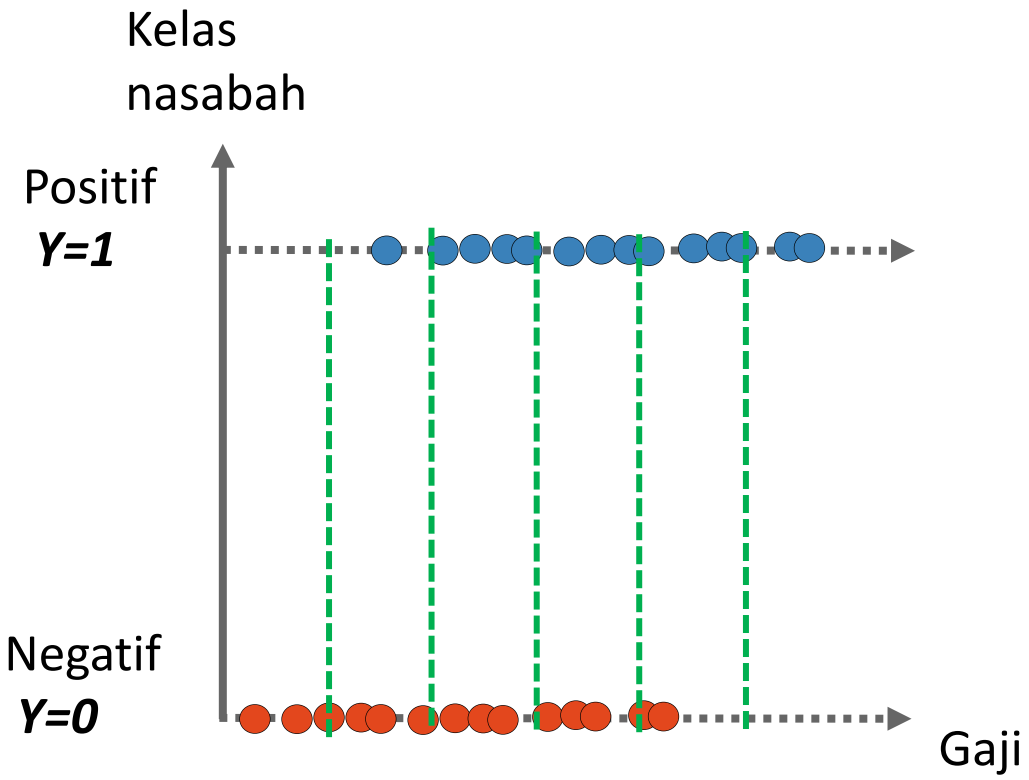
# Regresi Logistik: Permasalahan Umum




*Prediksi bisa tidak sesuai*

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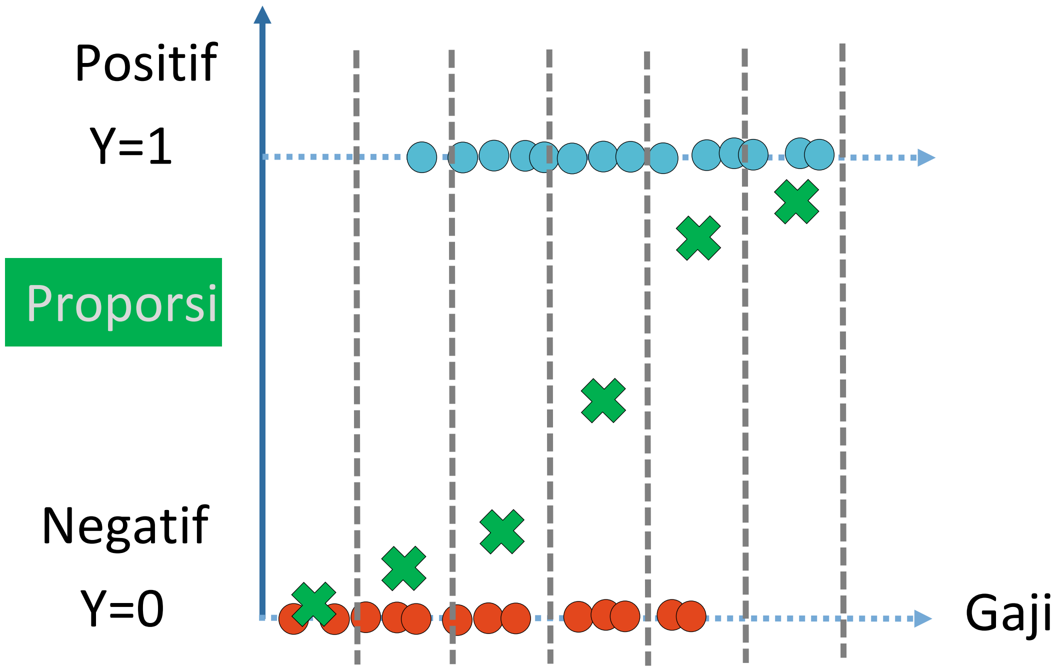
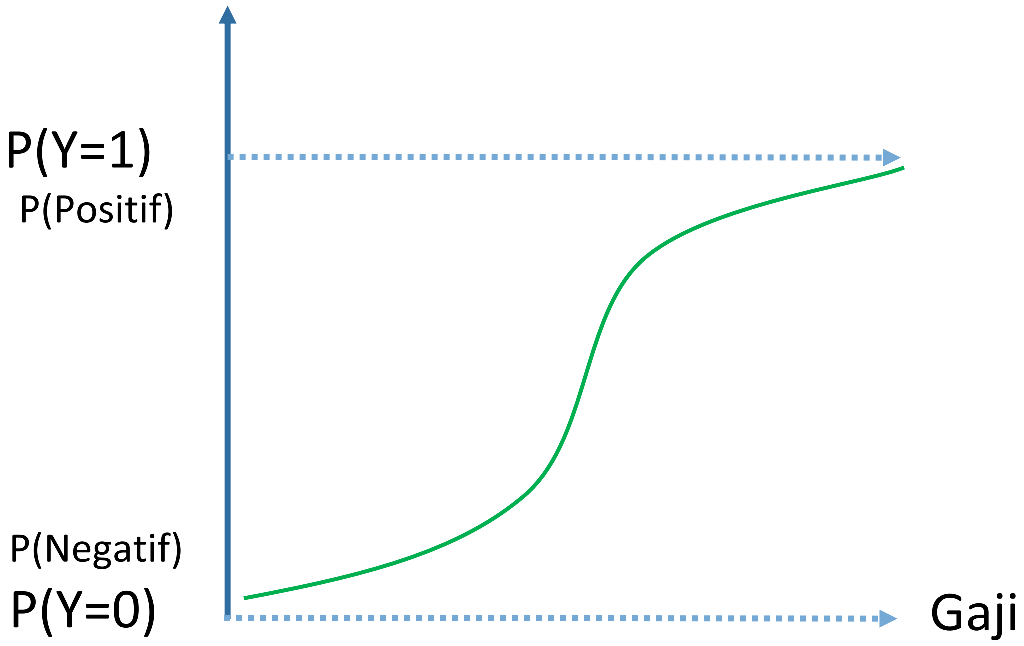
# Regresi Logistik: Permasalahan Umum




$$log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1$$

$$P(Y=1) = \frac{e^{\beta_0+\beta_1 X_1}}{1+e^{\beta_0+\beta_1 X_1}}$$

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# Konsep Dasar

$$log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1$$

$$P(Y = 1) = \frac{e^{\beta_0+\beta_1 X_1}}{1 + e^{\beta_0+\beta_1 X_1}}$$

Jika terdapat *p* variabel, maka secara umum dapat dituliskan sebagai berikut:

$$log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$

$$P(Y = 1) = \frac{e^{\beta_0+\beta_1 X_1+\ \beta_2 X_2+\cdots+\beta_p X_p}}{1 + e^{\beta_0+\beta_1 X_1+\ \beta_2 X_2+\cdots+\beta_p X_p}}$$

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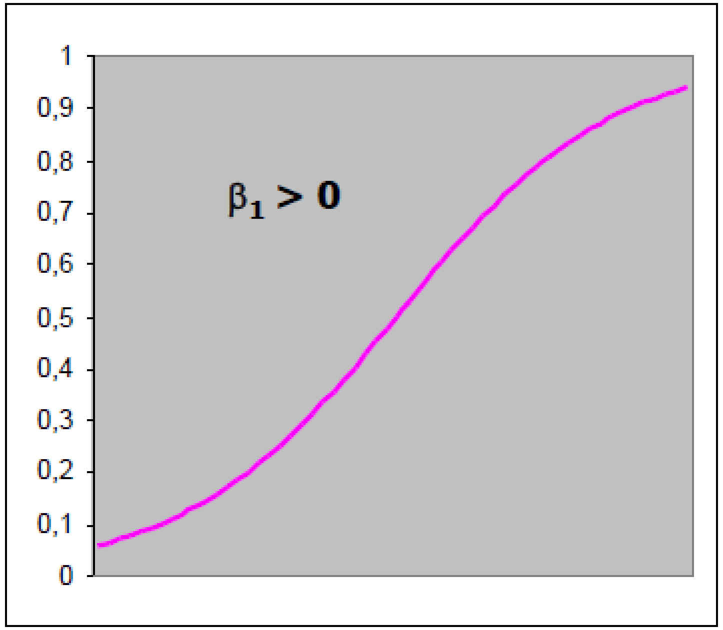
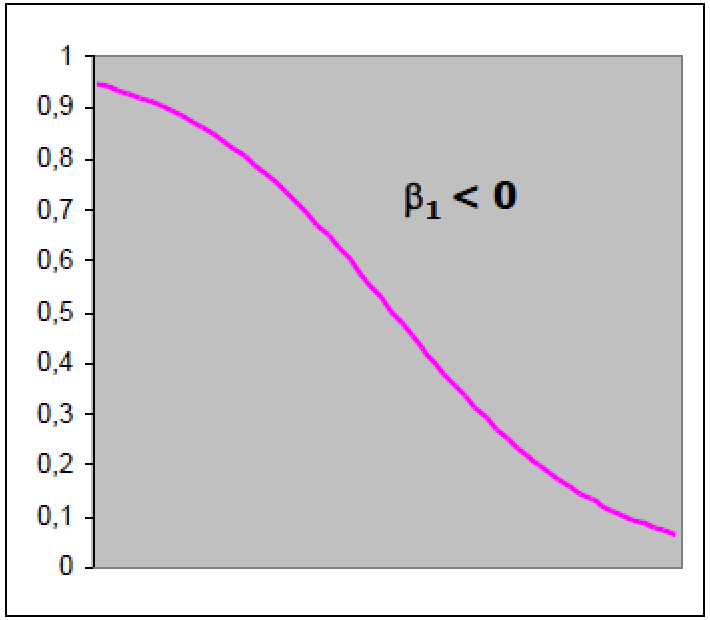
# 2. Model Regresi Logistik




*Nilai X bertambah maka P(Y=1) juga naik*




*Nilai X bertambah maka P(Y=1) turun*

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# 2 . Pendugaan Parameter

Fungsi *likelihood*

$$l(\boldsymbol{\beta}) = \prod_{i=1}^{n} \pi(x_i)^{y_i} \left[1 - \pi(x_i)\right]^{1-y_i}$$

Logaritma fungsi *likelihood*

$$L(\boldsymbol{\beta}) = \ln\left[l(\boldsymbol{\beta})\right] = \sum_{i=1}^{n} \left\{ y_i \ln\left[\pi(x_i)\right] + (1 - y_i)\ln\left[1 - \pi(x_i)\right] \right\}$$

Maksimum fungsi log-*likelihood* dengan syarat

$$\sum\left[y_i - \pi(x_i)\right] = 0$$ atau $$\sum x_i\left[y_i - \pi(x_i)\right] = 0$$

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# 2. Pengujian Model

## Uji Simultan

$H_0$: $\boldsymbol{\beta_1} = \boldsymbol{\beta_1} = \ldots = \boldsymbol{\beta_p} = 0$ (Model tanpa penjelas/interaksi)
$H_1$: $\boldsymbol{\beta_j} \neq 0$, $j = 1, 2, \ldots, p$ (Model dengan penjelas/interaksi)

$$D = -2\ln\left[\frac{\text{(likelihood of the fitted model)}}{\text{(likelihood of the saturated model)}}\right]$$ Deviance

$$G = -2\ln\left[\frac{\text{(likelihood without the variable)}}{\text{(likelihood with the variable)}}\right]$$

$$G = -2\ln\left[\text{(likelihood without the variable)} - \text{(likelihood with the variable)}\right].$$ Statistik G

*Likelihood ratio test statistics (G) ~ chi-square$_{(p)}$*

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# 2. Pengujian Model

**Uji Parsial**

H$_0$: $\beta_j = 0$, $j = 1, 2, \ldots, p$
H$_1$: $\beta_j \neq 0$

$$W_j = \frac{\hat{\beta}_j}{\widehat{\mathrm{SE}}(\hat{\beta}_j)}$$ Uji Wald

*Wald test statistics* (W) ~ Z

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# 3 . Interpretasi

***Odds***

Peluang sukses dibagi dibagi dengan peluang gagal:

$$odds = \frac{\pi}{1-\pi}$$

***Odds ratio***

Dugaan *odds* sukses sebesar OR kali $g_1$ dibanding $g_2$.

$$OR = \frac{\pi_1 / 1-\pi_1}{\pi_2 / 1-\pi_2} = \frac{\pi_1(1-\pi_2)}{\pi_2(1-\pi_1)}$$

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# Ilustrasi

Membeli: 1 = Ya, 0 = Tidak
Usia
Gender: 1 = Male, 0 = Female
Pemasukan: low, medium, high

| | income | age | gender | purchase |
|---|---|---|---|---|
| 1 | Low | 41 | Female | 0 |
| 2 | Low | 47 | Female | 0 |
| 3 | Low | 41 | Female | 1 |
| 4 | Low | 39 | Female | 1 |
| 5 | Low | 32 | Female | 0 |
| 6 | Low | 32 | Female | 0 |

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# Ilustrasi

```
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.28025    0.69605  -1.839   0.0659 .
age          0.01980    0.01761   1.125   0.2608
---
```

$$logit(\pi) = \ln\left(\frac{\pi}{1-\pi}\right) = \hat{\beta}_0 + \hat{\beta}_1 age$$

$$\widehat{OR} = e^{\widehat{\beta}_0 + \widehat{\beta}_1 age} = e^{\widehat{\beta}_0} e^{\widehat{\beta}_1}$$

$\hat{\beta}_0 = -1.28$, maka $e^{-1.28} = (2.718)^{-1.28} = 0.28$
maka dugaan odds pelanggan membeli ketika usia bernilai 0 adalah 0.28

$\hat{\beta}_1 = 0.02$ maka $e^{0.02} = (2.718)^{0.02} = 1.02$
maka dugaan odds pelanggan membeli bertambah sebesar 1.02 kali ketika usia bertambah (1 tahun)

# Ilustrasi

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.3194     0.1307  -2.442   0.0146 *
genderMale   -0.4373     0.2029  -2.155   0.0312 *
---
```

$$logit(\pi) = \ln\left(\frac{\pi}{1-\pi}\right) = \hat{\beta}_0 + \hat{\beta}_1 gender$$

$logit(\pi) = \ln\left(\frac{\pi}{1-\pi}\right) = \hat{\beta}_0 + \hat{\beta}_1(0)$, **Female**
$logit(\pi) = \ln\left(\frac{\pi}{1-\pi}\right) = \hat{\beta}_0 + \hat{\beta}_1(1)$, **Male**

$\widehat{OR} = e^{\hat{\beta}_0+\hat{\beta}_1(0)} = e^{\hat{\beta}_0}$, **Female**
$\widehat{OR} = e^{\hat{\beta}_0+\hat{\beta}_1(1)} = e^{\hat{\beta}_0}e^{\hat{\beta}_1}$, **Male**

$\boldsymbol{\hat{\beta}_0} = -0.32$, maka $e^{-0.32} = (2.718)^{-0.32} = 0.73$
maka dugaan odds pelanggan **Female** membeli sebesar 0.73.

$\boldsymbol{\hat{\beta}_1} = -0.44$ maka $e^{-0.44} = (2.718)^{-0.44} = 0.65$
maka dugaan odds pelanggan **Male** membeli sebesar 0.65 kali pelanggan **Female**

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# Evaluasi Model

| data training | | | |
|---|---|---|---|
| | budget | kesukaan | churn |
| 419 | medium | Seni | 1 |
| 847 | low | Tekno | 0 |
| 839 | low | Tekno | 0 |
| 400 | medium | Tekno | 1 |
| 486 | low | Tekno | 0 |
| 1006 | low | Busana | 0 |
| 356 | high | Busana | 0 |
| 876 | high | Tekno | 0 |
| 156 | low | Lainnya | 0 |
| 413 | low | Busana | 0 |

**Regresi logistik**

**validasi**

| data testing | | | |
|---|---|---|---|
| | budget | kesukaan | churn |
| 3 | low | Tekno | 0 |
| 8 | low | Seni | 0 |
| 10 | high | Tekno | 0 |
| 13 | medium | Tekno | 1 |

| Prediksi P (Y = 1) |
|---|
| 0.43 |
| 0.84 |
| 0.22 |
| 0.53 |

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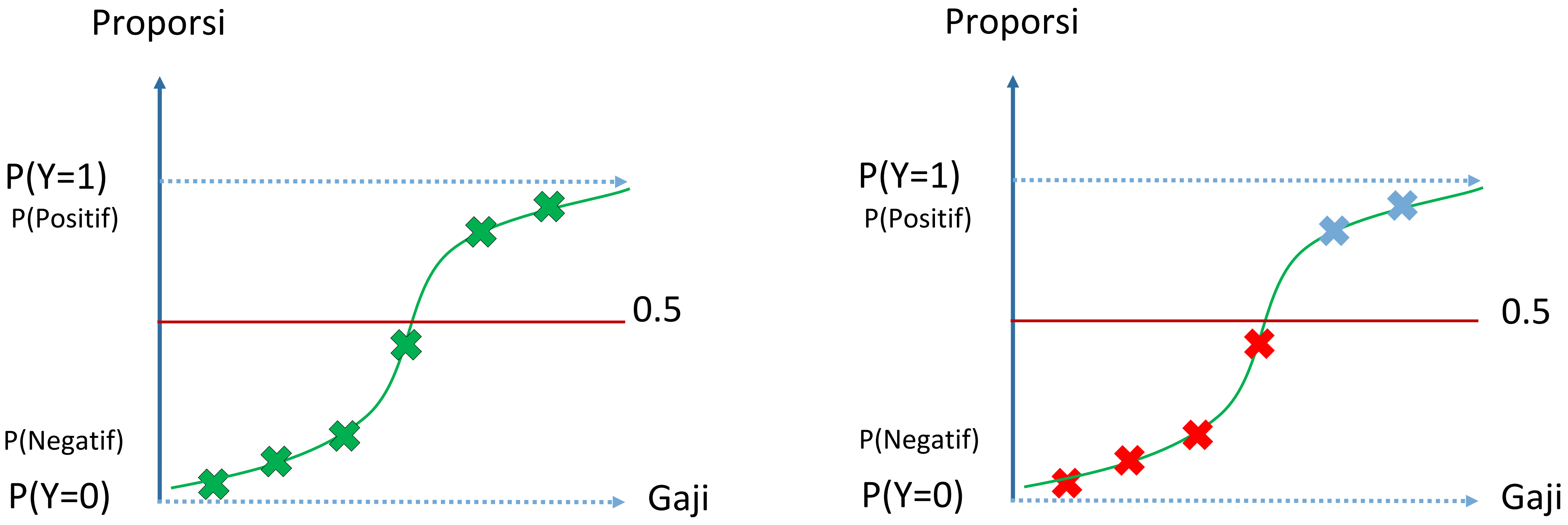
# Evaluasi Model

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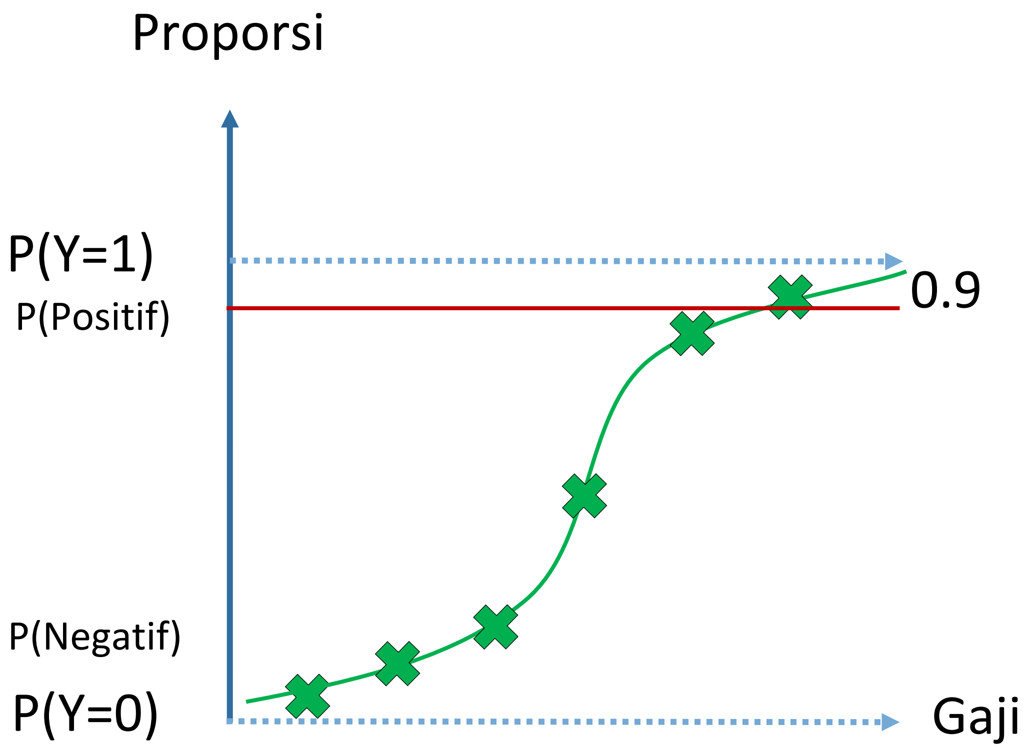


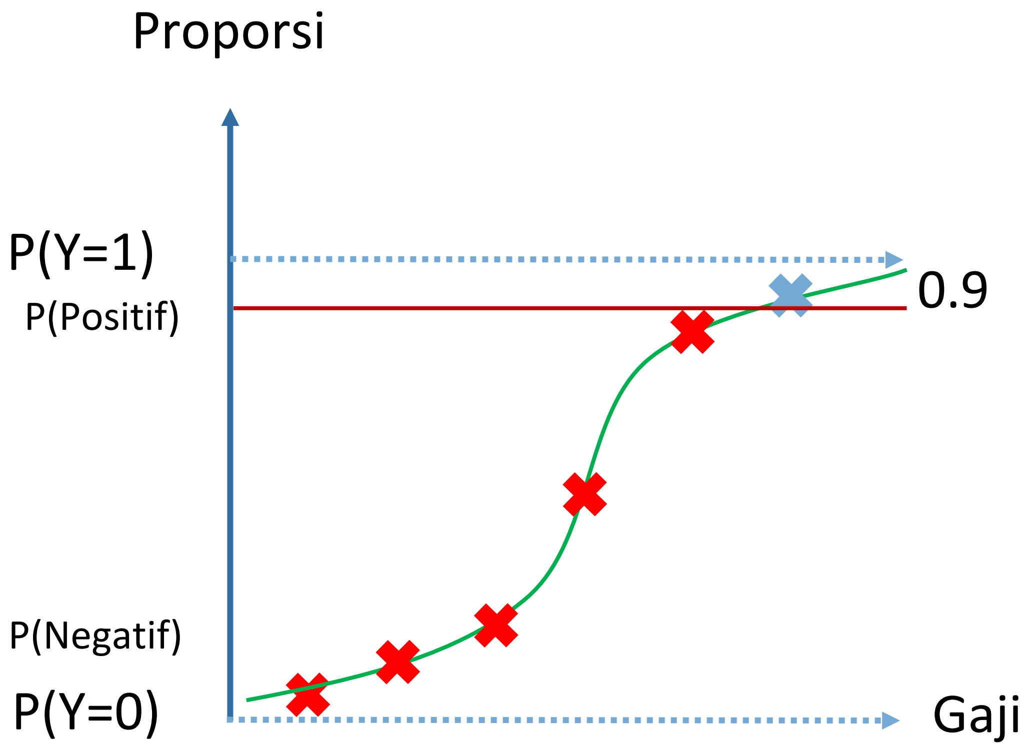


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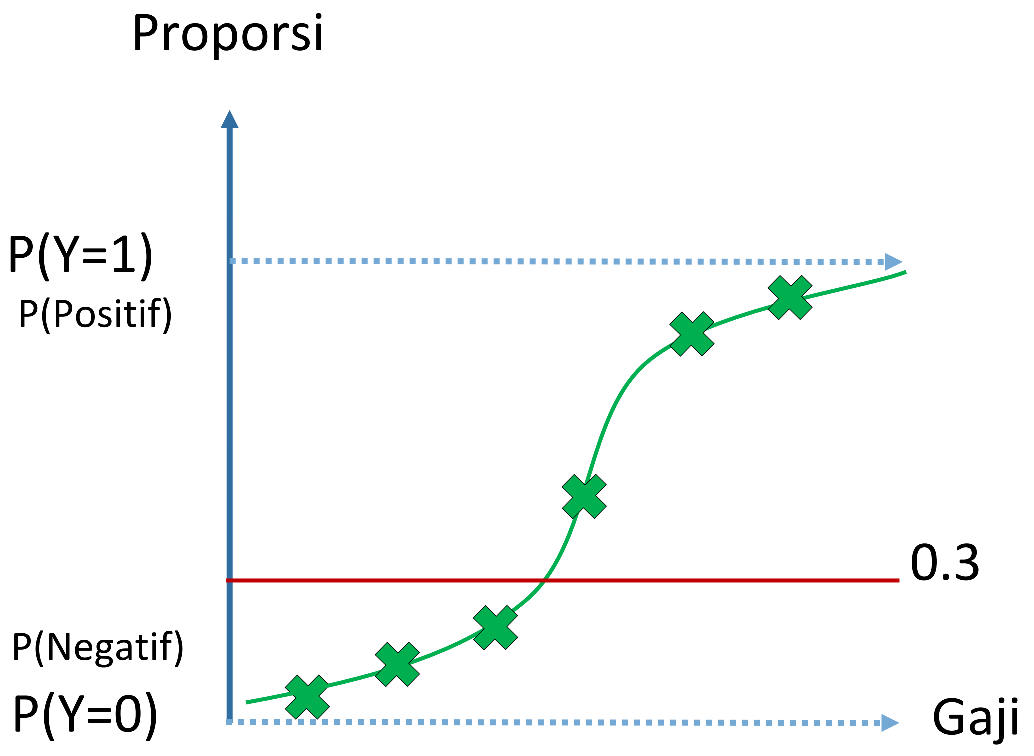


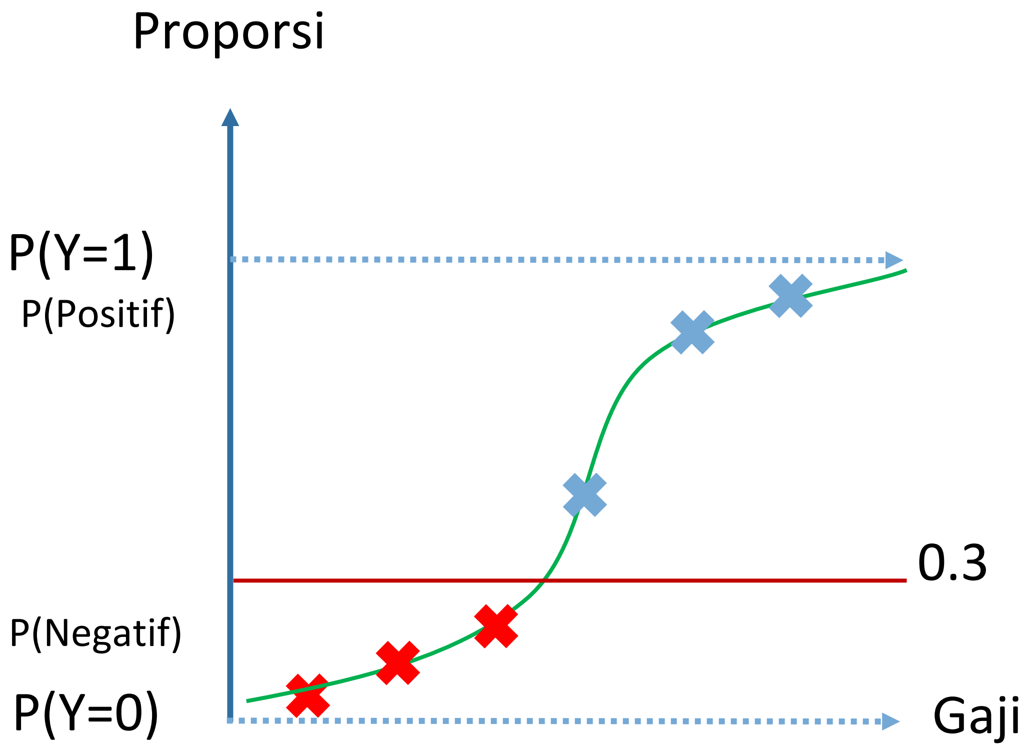


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# Evaluasi Model

**data training**

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| 419 | medium | Seni | 1 |
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**Regresi logistik**

**validasi**

**data testing**

| | budget | kesukaan | churn |
|---|---|---|---|
| 3 | low | Tekno | 0 |
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*cut-off* 0.5

| Prediksi P (Y = 1) |
|---|
| 0.43 |
| 0.84 |
| 0.22 |
| 0.53 |

| Prediksi P (Y = 1) |
|---|
| 0 |
| 1 |
| 0 |
| 1 |

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# Evaluasi Model

Tabel klasifikasi (*confusion matrix*)

| | | **Prediksi kelas** | |
|---|---|---|---|
| | | Positif | Negatif |
| **Kelas sebenarnya** | Positif | TP | FN |
| | Negatif | FP | TN |

$$Akurasi = \frac{TP + TN}{TP + FP + FN + TN}$$

$$\underset{recall}{Sensitifitas} = \frac{TP}{TP + FN}$$

$$Spesifisitas = \frac{TN}{TN + FP}$$

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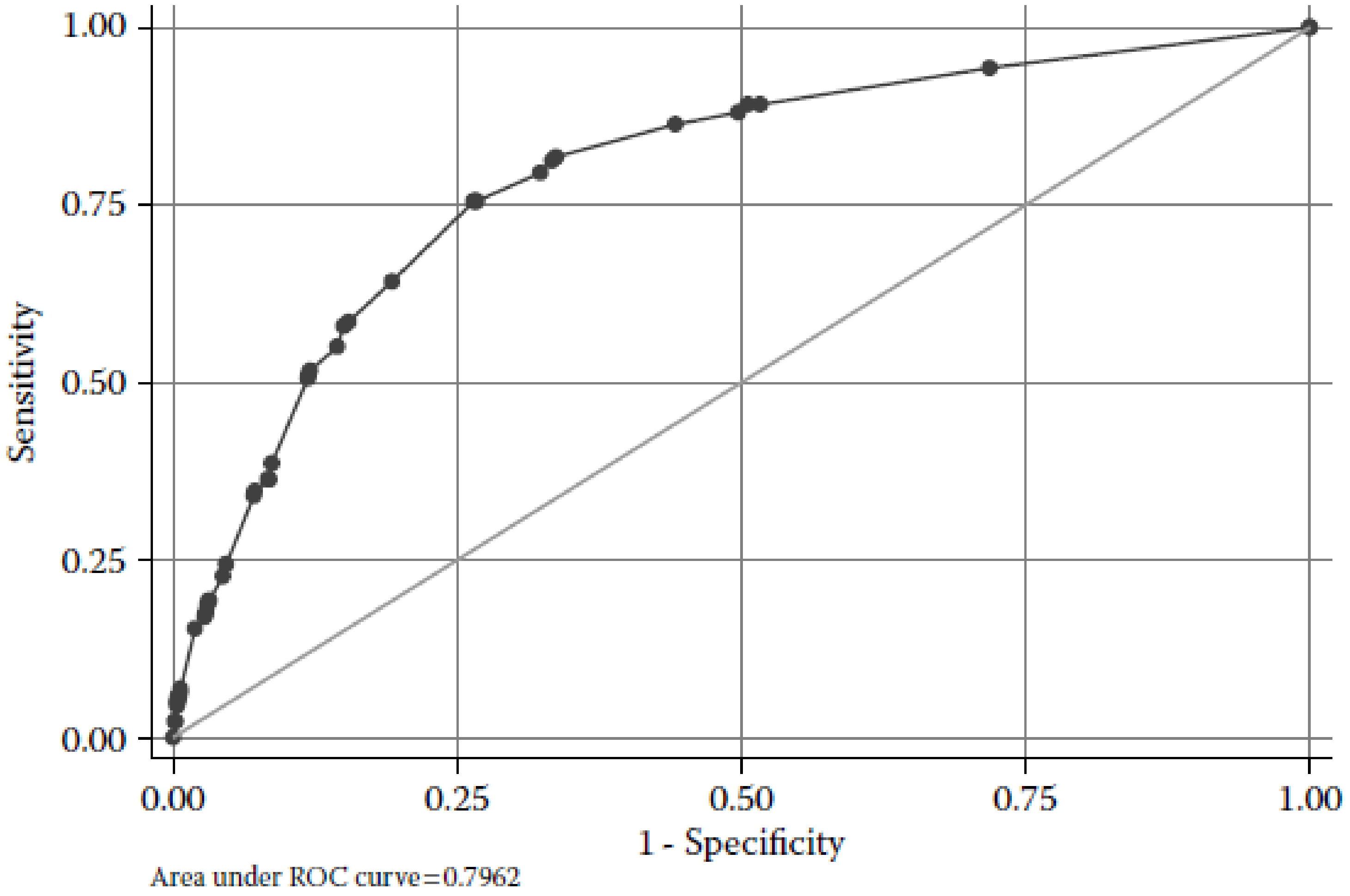
# Evaluasi Model




| ROC-AUC | Akurasi |
|---|---|
| 0.5 – 0.6 | Poor |
| 0.6 – 0.7 | Average |
| 0.7 – 0.8 | Good |
| 0.8 – 0.9 | Very good |
| 0.9 – 1.0 | Excellent |

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Semoga bermanfaat
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